

DRIFT, TRACKING AND THE STABILITY OF VORTEX TUBES IN TEARING STRAIN

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ABSTRACT. In the exact reduction of a vortex tube in a uniform strain, the first moment of vorticity obeys $\frac{d}{dt} \int x\omega = D \int x\omega$ exactly, for any sign structure, and the equation is invariant under shifts along solutions of $\dot{\xi} = D\xi$. So in the tearing regime ($\lambda > 1$) a tube drifts off its stagnation point at the rate $s(\lambda - 1)/2$, whatever its structure. We ask whether surrounding vortices can hold it. For relative equilibria of same-sign point vortices we prove they cannot: after a $\sqrt{\Gamma}$ rescaling the linearisation is complex symmetric, so the coupled drift rate is $\sqrt{s_{\max}(E)^2 - \Omega^2} \geq \sqrt{|E_{cc}|^2 - \Omega^2}$, the rate with the others frozen. Opposite-sign partners break the symmetry and can hold the core in a neutral dipole, which is what we observe in ensembles. For time-dependent same-sign supply the inequality fails pointwise but holds in the median. In three dimensions short axial waves are held by self-induction and stretched into the planar drift (no 3D instability at $\lambda = 2$), the elliptic and curvature instabilities give bounded gains, and tilting maps $\lambda > 3$ into $(1, 3)$. The self-induced strain at a Gaussian core's centre reinforces the external strain, by a ratio tending to $3/2$.

1. INTRODUCTION

A vortex tube in a uniform strain with axial stretching s and in-plane strain difference λs sits at the stagnation point of the strain. For $\lambda < 1$ that point is a sink in the plane and the tube settles into a non-symmetric Burgers vortex [6, 3]. For $\lambda > 1$ one in-plane direction is extensional at rate $s(\lambda - 1)/2$, the stagnation point is a saddle, and the tube both leaks circulation, exponentially slowly [8, 9], and drifts off the saddle. This paper is about the drift.

Section 2 shows that the drift is exact and structure-independent: the centroid of any vorticity distribution moves as a passive point. Whether the tube can still be held then depends on the *other* vortices around it, which respond to its displacement. Section 3 proves that same-sign vortices in a relative equilibrium never reduce the drift rate (Theorem 3.1), by a complex-symmetric (Takagi) structure that opposite signs destroy. Section 4 tests the time-dependent case numerically: the theorem fails pointwise but holds in distribution, and the cases where the drift is suppressed are dipole formation. Section 5 collects the three-dimensional effects: axial waves, the elliptic and curvature instabilities, and tilting. Section 6 records a fact about the strain a core induces at its own centre, with a consequence for Miller's middle-eigenvalue criterion [5].

2. THE EXACT DRIFT

For a straight tube along z in the strain $(-g_1x, -g_2y, sz)$ with $g_1 = \frac{1+\lambda}{2}s$, $g_2 = \frac{1-\lambda}{2}s$, the vorticity $\omega(x, y, t)e_z$ obeys

$$\partial_t \omega + \nabla \cdot ((Dx + u_\omega) \omega) = \nu \Delta \omega, \quad D = -\text{diag}(g_1, g_2), \quad u_\omega = K * \omega, \quad K(x) = \frac{x^\perp}{2\pi|x|^2}. \quad (1)$$

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Proposition 2.1 (exact drift). *For every solution of (1) with enough decay, of any sign,*

$$\frac{d}{dt} \int x \omega dx = D \int x \omega dx, \quad \frac{d}{dt} \int \omega dx = 0.$$

If $\omega(x, t)$ is a solution, so is $\omega(x - \xi(t), t)$ for every solution of $\dot{\xi} = D\xi$. In particular, for $\Gamma = \int \omega \neq 0$ the centroid $x_c = \Gamma^{-1} \int x \omega$ obeys $\dot{x}_c = Dx_c$ exactly, and position decouples from shape.

Proof. Multiply (1) by x and integrate by parts: $\int x \nabla \cdot (Dx \omega) = - \int Dx \omega$, $\int x \Delta \omega = 0$, and $\int u_\omega \omega = \iint K(x-y) \omega(x) \omega(y) dx dy = 0$ because K is odd. For the shift, put $\tilde{\omega}(x, t) = \omega(x - \xi, t)$, so $u_{\tilde{\omega}}(x) = u_\omega(x - \xi)$ and $Dx = D(x - \xi) + D\xi$. Then $\partial_t \tilde{\omega} = (\partial_t \omega)(x - \xi) - \dot{\xi} \cdot \nabla \tilde{\omega}$, and $\nabla \cdot (D\xi \tilde{\omega}) = D\xi \cdot \nabla \tilde{\omega}$; the two extra terms cancel exactly when $\dot{\xi} = D\xi$. $\square$

Remark 2.2. (a) For $\lambda > 1$ the eigenvalue $-g_2 = s(\lambda - 1)/2$ of D is positive: any offset of the centroid from the stagnation point grows like $e^{s(\lambda-1)t/2}$, independent of the tube's internal structure, circulation and viscosity. (b) For $\Gamma = 0$ the first moment of a dipole, $\Gamma_+ d$, obeys the same law: its separation is pulled apart along the outflow axis or pushed together into cancellation along the inflow axis. (c) In the tube's own frame this is a frame effect; it matters whenever the strain is set by other structures, which is the situation of a collapse fed by its surroundings.

Check. In two-dimensional simulations of (1) at $\lambda = 3$ the offset grows at the predicted rate to within 1%. Without symmetry enforcement a tube at $\lambda = 4$, $\Gamma/\nu = 1000$ is displaced by round-off and leaves a measuring disc of radius 4 at $t_{1/2} = 23.83$ on $[-20, 20]^2$ and 23.97 on $[-12, 12]^2$, against $\ln(10^{16})/|g_2| \approx 24.6$: the ‘‘abrupt tearing’’ of such runs is this drift.

3. SUPPLY NEVER TRACKS AT EQUILIBRIUM

Suppose the strain on a core comes from other vortices. Displacing the core displaces the strain field too, because the other vortices respond. Define the *blind rate* σ_{blind} as the growth rate of a core displacement with the other vortices held on their unperturbed motion, and the *full rate* σ_{full} as the same with all vortices responding. The response *tracks* the core if $\Lambda = \sigma_{\text{full}}/\sigma_{\text{blind}} < 1$.

Theorem 3.1 (supply never tracks at equilibrium). *Let $z_1, \dots, z_N$ be a relative equilibrium of point vortices rotating at Ω , with all $\Gamma_j > 0$. For every vortex c ,*

$$\sigma_{\text{full}} = \sqrt{s_{\text{max}}(E)^2 - \Omega^2} \geq \sqrt{|E_{cc}|^2 - \Omega^2} = \sigma_{\text{blind}}(c)$$

(real parts, with the convention $\sqrt{\cdot} = 0$ for negative arguments), where E is the complex-symmetric matrix below and s_{max} its largest singular value. The inequality is strict whenever c interacts with another vortex.

Proof. The conjugate velocity $\bar{v}_j = \frac{1}{2\pi i} \sum_{k \neq j} \Gamma_k / (z_j - z_k)$ is holomorphic in the positions. In the rotating frame a perturbation w obeys $\dot{w} = -i\Omega w + F\bar{w}$ with $F_{jk} = -\Gamma_k \bar{c}_{jk}$ for $k \neq j$, $F_{jj} = \sum_{k \neq j} \Gamma_k \bar{c}_{jk}$, and $c_{jk} = c_{kj} = -1/(2\pi i(z_j - z_k)^2)$. Put $u_j = \sqrt{\Gamma_j} w_j$, which is a real rescaling because $\Gamma_j > 0$. Then $\dot{u} = -i\Omega u + E\bar{u}$ with $E = DFD^{-1}$, $D = \text{diag}(\sqrt{\Gamma_j})$, and $E_{jk} = -\sqrt{\Gamma_j \Gamma_k} \bar{c}_{jk} = E_{kj}$: E is complex symmetric. For a real exponent σ , $(\sigma + i\Omega)u = E\bar{u}$; conjugating and substituting, $(\sigma^2 + \Omega^2)u = E\bar{E}u = EE^*u$, so $\sigma^2 + \Omega^2$ is a squared singular value of E . Conversely, Takagi's factorisation $E = U\Sigma U^T$ [4] gives vectors with $E\bar{u}_k = s_k u_k$, and $u = \alpha u_k$ with $|\alpha| = 1$, $\alpha^2 = s_k/(\sigma + i\Omega)$ solves the equation whenever $s_k > |\Omega|$. Hence $\sigma_{\text{full}} = \sqrt{s_{\text{max}}^2 - \Omega^2}$. The blind system is the 1×1 case, $\sigma_{\text{blind}} = \sqrt{|E_{cc}|^2 - \Omega^2}$. Finally $s_{\text{max}} = \|E\| \geq \|Ee_c\| \geq |E_{cc}|$, with strict inequality if column c has another nonzero entry. $\square$

Remark 3.2 (structure). With $E = P + iQ$ the linearisation in real form is $\frac{d}{dt}(a, b) = \begin{pmatrix} P & Q \\ Q & -P \end{pmatrix} (a, b) - \Omega \mathbb{J}(a, b)$: real symmetric and traceless, plus the frame rotation. The theorem is Cauchy interlacing in disguise, which has no time-dependent analogue (Section 4). With some $\Gamma_k < 0$ the rescaling uses $\sqrt{|\Gamma_k|}$, $E_{jk} = -\text{sgn}(\Gamma_k) \sqrt{|\Gamma_j \Gamma_k|} \bar{c}_{jk}$ is no longer symmetric, $E\bar{E}$ is not Hermitian, and nothing bounds σ_{full} below by the blind rate.

Numerical checks. Over 726 hyperbolic (vortex, equilibrium) pairs with $N = 3$ –6 same-sign vortices, $\Lambda \geq 1.09$ throughout, and σ_{full} equals $\sqrt{s_{\text{max}}^2 - \Omega^2}$ to 10^{-5} on 138 equilibria. Examples:

- the symmetric collinear triple (core Γ_c between two suppliers Γ_s at $\pm d$, hyperbolic iff $\Gamma_c < 1.5\Gamma_s$): $\Lambda = 1.03, 1.06, 1.11, 1.18, 1.32, 1.51, 1.78, 1.96, 2.21, 2.57, 3.17, 4.52, 6.40$ for $\Gamma_c/\Gamma_s = 0.05$ to 1.45 . The blind rate falls from 0.306 to 0.071 as the frame rotation grows, while the full rate rises from 0.314 to 0.453 : the response *amplifies* the drift. Co-rotating N -gons with $N \geq 3$ have no hyperbolic centre at all;
- asymmetric co-rotating triples $\Lambda = 1.09$ – 5.05 ; co-rotating five-chains 1.23 – 1.57 ;
- counter-rotating control $(-\Gamma_s, \Gamma_c, -\Gamma_s)$: $\Lambda = 0.95, 0.85, 0.75, 0.65, 0.52$ for $\Gamma_c/\Gamma_s = 0.1$ to 0.9 ; the drift is slowed but never cancelled.

Finite cores. Two-dimensional Navier–Stokes with three Gaussian vortices ($\Gamma_s = 50$, $\Gamma_s/\nu = 5000$, core radius 0.5 , spacing 3) gives drift rates $2.63, 2.75, 2.78$ at $\Gamma_c/\Gamma_s = 0.5, 1.0, 1.3$, against the point-vortex full rates $2.03, 2.30, 2.45$ and blind rates $1.53, 1.17, 0.77$: the amplification survives finite viscous cores and is slightly stronger.

4. TIME-DEPENDENT SUPPLY

For configurations that are not equilibria the rates are finite-time Lyapunov rates. We compute them tangent-linearly along the reference run ($\dot{w} = F(t)\bar{w}$ for the full rate, $\dot{w}_c = F_{cc}(t)\bar{w}_c$ for the blind one), which avoids the saturation of finite- δ estimates. After the $\sqrt{\Gamma}$ rescaling the matrix is still complex symmetric at every instant; only the rotation of its Takagi vectors can break the comparison.

Free ensembles (core plus 4–6 vortices in $1 \leq |z| \leq 2$, horizon $10/e_0$): with all signs equal, median $\Lambda = 2.06$, 21 of 142 below 1, 3 below 0.5; with random signs, median 0.70 , 99 of 143 below 1, 46 below 0.5. Bound co-rotating triples track in the Lyapunov sense at long times, with the core in strain that rotates with the cluster.

In a fixed external strain (the setting of a tube fed from larger scales): point vortices in the imposed field $-\frac{s}{2}z - e\bar{z}$, $\lambda \in \{1.5, 2, 3\}$, the core at the saddle, 3–6 same-sign suppliers arriving along the inflow axis; runs stop at the first close approach (for point vortices, the merger). Where the imposed strain dominates the suppliers' (ratio $\rho < 1$, 86 of 300 configurations): median $\Lambda = 1.02$, minimum 0.64 , 4 below 0.99 (three of them marginal). Where the suppliers dominate ($\rho \geq 1$): median 1.12 , 84 of 214 below 0.99 . A mixed-sign control has 21 of 52 strain-dominated cases below 0.99 : suppression is about eight times more frequent with opposite-sign vorticity present.

What the suppressing cases are. In the free mixed-sign ensembles, every case with $\Lambda < 0.3$ at $10/e_0$ has one dominant *counter-rotating* partner: the same vortex is nearest 87–100% of the time in 11 of 13 cases and supplies a median 79% of the core's strain, at a moderate, persistent distance. The core forms a neutrally stable pair that translates; it is not held at a saddle, and its partner adds no circulation.

Remark 4.1. So the time-dependent version of Theorem 3.1 is false pointwise even for same-sign supply in a fixed-axis strain, but true in the median of every subset tested. Suppressing the

drift for long needs opposite-sign vorticity near the core, which delivers strain but removes circulation.

5. THREE DIMENSIONS

Axial waves. Perturb the equilibria of Section 3, now straight filaments of core radius $a = 0.1d$, by $\cos kz$. Mutual induction is given in closed form by $\delta u = \frac{\Gamma}{4\pi} \left[-\hat{z} \times \eta \left(\frac{2kK_1(kR)}{R} + 2k^2 K_0(kR) \right) + \hat{z} \times R (R \cdot \eta) \frac{2k^2 K_2(kR)}{R^2} \right]$, which reduces to the Kirchhoff Jacobian as $k \rightarrow 0$, and self-induction by the retrograde rotation $\frac{\Gamma k^2}{4\pi} \left(\ln \frac{2}{ka} - \gamma + \frac{1}{4} \right)$. Over ten equilibria and $kd \in (0, 5]$ no case cancels the drift (worst $\Lambda = 0.52$, counter-rotating). For $kd \gtrsim 1$ –2 the blind rate vanishes: the core's self-induced Kelvin-wave rotation holds it against short-wavelength displacements by itself. The drift is a long-wave mode.

A 3D simulation. We solve the full vorticity equation for a tube in the strain at $\lambda = 2$, $\Gamma/\nu = 419$ ($\varepsilon = 0.03$), absorbing the stretching in the coordinate $\zeta = ze^{-st}$ (periodic, period 16π , 32 points), so axial mode m has wavenumber $\frac{m}{8}e^{-st}$; in the plane, free space $[-12, 12]^2$ at 128^2 , velocity from the vector potential per axial mode. The initial quasi-steady tube is displaced along the outflow axis by $0.05 \sum_m \cos(mk_1\zeta + \phi_m)$, $m \in \{0, 1, 2, 4, 8\}$. Circulation is conserved to 0.3% over 6/s. Registered predictions and results: the $m = 0$ mode grows at 0.491 (predicted 0.50 ± 0.10); on $t \in [4, 6]$ every mode grows at 0.460–0.475 (predicted 0.50 ± 0.15); no mode sustains a rate above 0.505 over any unit window (a genuine 3D instability would exceed 0.75). The short modes are first held and decay ($m = 4$: $0.045 \rightarrow 0.011$; $m = 8$: $0.045 \rightarrow 0.007$), then grow at the drift rate once stretching has made them long. The predicted release order failed ($m = 8$ before $m = 4$), a defect of the Rankine cut-off in the estimate, not of the picture. At these parameters the planar reduction (drift and leak) is the right model in 3D, and short axial waves are more protected than the planar tube.

Elliptic instability. Eloy and Le Dizès [1] give, for Burgers vortices in a perpendicular strain, a transient gain above 10^4 only if $e/s > 0.2(\Gamma/\nu)^{1/2}$, i.e. $\Gamma/\nu < 6.25\lambda^2$. Tubes with ε below the survival threshold of [9] have $\Gamma/\nu \gtrsim 125\lambda$, so they are stable to it for $\lambda < 20$. A WKB scan with axial stretching of the quasi-steady vortices at $\varepsilon = 0.030$ –0.015 gives a maximal viscous gain $e^{1.9}$ over $t \leq 10/s$.

Curvature instability. The curvature instability of rings (Fukumoto and Hattori [2]) grows at $\sigma_c = C \varepsilon_c \omega_0$ with $\varepsilon_c = a/R_c$. A WKB computation on relaxed viscous rings gives $C = 0.147, 0.157, 0.168$ at $\varepsilon_c = 0.10, 0.15, 0.20$: the Gaussian-core constant is ≈ 0.157 , about a quarter of the Rankine value 0.645. Axial stretching both straightens the ring and detunes the resonance. On a stretched Burgers ring with $\varepsilon_c(0) = 0.15$ the maximal $\ln G$ over $st \leq 1.5$ is 0.64 ($\Gamma/\nu = 300$) and 1.13 (1000), against 1.77 and 4.32 for unstretched controls and the straightening bounds 2.7 and 9.2. Detuning does most of the work; the gain stays below a factor 3.1.

Tilting. For a tube aligned with a strain eigenvector the parameter λ is a function of the eigenvalues, which gives the following.

Lemma 5.1 (eigenvalue identity). *Let the external strain have eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3$, $\sum \lambda_i = 0$. For a tube along the λ_3 eigenvector, $\lambda = (\lambda_2 - \lambda_1)/\lambda_3$, and $\lambda > 1 \iff \lambda_2 > 0$; always $\lambda < 3$. For a tube along the λ_2 eigenvector with $\lambda_2 > 0$, $\lambda = 1 + 2\lambda_3/\lambda_2 > 1$, and $\lambda > 3 \iff \lambda_3 > \lambda_2$. Realigning such a tube with λ_3 gives*

$$\lambda' = \frac{\lambda + 3}{\lambda - 1} \in (1, 3), \quad s' = \frac{s(\lambda - 1)}{2}.$$

Proof. On the λ_3 axis, $\lambda = (\lambda_2 - \lambda_1)/\lambda_3 = (2\lambda_2 + \lambda_3)/\lambda_3$, which exceeds 1 iff $\lambda_2 > 0$ and is below 3 iff $\lambda_2 < \lambda_3$. On the λ_2 axis, $\lambda = (\lambda_3 - \lambda_1)/\lambda_2 = (\lambda_2 + 2\lambda_3)/\lambda_2$. Eliminating $\lambda_3/\lambda_2 = (\lambda - 1)/2$ gives $\lambda' = 1 + 2\lambda_2/\lambda_3 = (\lambda + 3)/(\lambda - 1)$ and $s' = \lambda_3 = s(\lambda - 1)/2$ with $s = \lambda_2$. $\square$

Prochazka and Pullin [8] show that a tube with $\lambda > 3$ tilts towards the most extensional direction at a rate of order $(\lambda_3 - \lambda_2)$, after which “a situation corresponding to $1 < \lambda < 3$ will prevail”. Lemma 5.1 identifies the endpoint. *Estimate.* From a misalignment θ_0 the realignment takes $t_{\text{tilt}} \approx \ln(\pi/2\theta_0)/(\lambda_3 - \lambda_2)$, during which the tube is in the tearing regime and leaks at rate κs of [9]; the loss is $O(\kappa \ln(1/\theta_0))$, exponentially small for strong tubes. Tilting does not take a tube out of the tearing regime; it moves it into $1 < \lambda \leq 3$.

6. SELF-STRAIN AT THE CORE

Proposition 6.1 (numerical). *At the centroid of a quasi-steady core in an external strain, the strain induced by the core’s own vorticity is aligned with the external strain and reinforces it. The ratio $K = e_{\text{self}}/e_{\text{ext}}$ rises with circulation to 1.50 at every λ tested: $K = 0.07, 0.52, 1.38, 1.50$ at $\Gamma/\nu = 10, 30, 100, 300$ for $\lambda = 0.5$ (steady non-symmetric Burgers vortices); likewise at $\lambda = 0.25, 0.75$; and $K = 1.53\text{--}1.63$ at $\lambda = 2$ (quasi-steady, $\varepsilon = 0.015\text{--}0.04$).*

For a uniform elliptical patch in weak strain (Moore–Saffman [7]) the ratio is 1; K depends on the profile. A derivation of the Gaussian limit $3/2$ from the large-circulation asymptotics of [6] is open.

Remark 6.2 (Miller’s criterion). Miller [5] shows that blowup requires λ_2^+ of the total strain to be large. By Lemma 5.1 the external strain on an aligned tube has $\lambda_2 > 0$ iff the tube is in the tearing regime. Proposition 6.1 shows that this does not transfer from total to external strain: at a tube centre the total λ_2 is positive as soon as $\lambda_{\text{ext}} > 1/(1 + K) \approx 0.4$, and off-centre any intense vortex has $\lambda_2^+ > 0$ from its own swirl. Miller’s criterion therefore does not force a collapse into the tearing regime; protected tubes are compatible with it.

7. OPEN PROBLEMS

Gap 7.1 (continuous vorticity). The holomorphic structure behind Theorem 3.1 persists for the linearised Euler coupling of a core to a continuous same-sign field, with $\sqrt{\omega}$ weights; the analogue of E is a complex-symmetric integral operator. A statement for steady same-sign vortex configurations should follow; it is not written.

Gap 7.2 (time-dependent supply). A distributional version of Theorem 3.1: bounds on how long, and how often, strong same-sign arrival can stall the drift. Whether stalls can be chained at every scale is the question a collapse fed by its surroundings would have to answer.

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